Syed Zulqarnain Haider *Electrical Engineering Graduate*

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zulkrnen.github.io

Profile

Electrical Engineering graduate with experience across technical projects, applied research, and software and hardware development. Combines a broad technical foundation with a **systems-oriented, cross-disciplinary approach to problem-solving, conceptualization, and practical implementation**. Enjoys exploring how technology can be applied to address real-world needs and improve the way systems work.

Education

Matriculation 2017 – 2018
Federal Board of Intermediate and Secondary Education Skardu, GB
Scored **94.9%**

FSC Pre-Engineering 2019 – 2020
Federal Board of Intermediate and Secondary Education Skardu, GB
Scored **87.9%**

BS Electrical Engineering 12/2021 – 12/2025
Institute of Electrical Electronics and Computer Engineering Lahore
IEECE - University of the Punjab
CGPA - **3.36/4.00**
Final-Year Design Project (FYDP) / Thesis - *Brainwave Signal Analysis Using AI to Evaluate Cognitive and Emotional Responses in Children Aged 2 - 7 to Visual and Auditory Stimuli.*

- **Project -** *NeuroSense*: *a web app and DL models developed and trained as part of the project.*

Research & Professional Experience

Control System Study 03/2024 – 07/2024
Intelligent Systems Lab & Automation Facility - ISLAF, PU Lahore

- **Extended a Control Systems course project into a research study** on state-feedback control for synchronizing two motors, with an aero-pendulum as the intended physical application.
- **Modeled the system and performed control-design calculations in MATLAB**, designing a state-feedback controller and an op-amp 741-based circuit in Multisim, then implementing the circuit on a breadboard and testing it using Feedback Instruments servo trainers.
- **Troubleshot an initial physical prototype** and adapted the experimental approach when local fabrication and precision-tooling constraints hindered mechanical and sensor integration, transitioning to laboratory servo trainers for further implementation and validation.

- Contributed to the **IEEE research paper, technical poster, and experimental documentation**, incorporating revisions under faculty supervision. Independently learned **LaTeX** to format the manuscript for publication.

Related publication: *Design & Analog Implementation of Synchronization Controllers for Feedback Instruments' Servo Trainer Module* — IEEE. DOI: 10.1109/ICIT63607.2024.10860018

Network Testing Internship 06/2024 – 08/2024
LCC Pakistan Lahore

- **Supported optimization of field-testing routes** within assigned coverage areas to improve route efficiency and minimize wasted travel time and fuel while assessing mobile network performance across voice calls, messaging, and mobile data services.
- **Operated drive-testing tools** to collect, organize, and document field data and observations for review by the technical team.
- **Participated in the on-site investigation of a client-reported connectivity issue**, helping identify a likely antenna-orientation issue and communicating the findings to the supervisor for follow-up by the designated implementation team.
- Developed practical familiarity with network testing procedures and field equipment through **team-based work and independent learning.**

Relevant Projects

NeuroSense: The Brain IMDb - Web App & DL Model
Final Year Design Project | University of the Punjab

- **Took primary responsibility for the project's research and technical execution**, developing an EEG-based system to analyze children's cognitive and emotional responses to visual and auditory stimuli.
- **Helped design and refine the study methodology**, conducting a pilot study with children and revising the data-collection procedure based on psychologist input, observed behaviour, and recording constraints before proceeding with school-based data collection.
- **Led most of the EEG data-processing and model-development work**, including data cleaning, stimulus-timeline annotation, dataset preparation, and experimentation with classical machine-learning models and recurrent deep-learning architectures and iterative hyperparameter and architecture tuning.
- **Addressed data imbalance and model generalization challenges** through data augmentation, additional data-processing experiments, and refinement of the classification categories to improve test performance.
- **Developed and deployed the NeuroSense demonstration web application on Hugging Face**, providing an interface for presenting the project's model-based analysis.
- **Prepared the project's key research documentation**, including successive proposal revisions, presentation materials, the thesis, and the research paper draft.

Multi-Campus School Management System
Software Development Hobby Project

- Designed, developed, and deployed a web-based school management system intended to support administrative and academic operations across multiple campuses.

- Worked across the development process, translating system requirements into an implemented application.

Simplified Network Design for a Small Corporate Office
Computer Networks - Academic Project
- **Engineered a simulated network infrastructure**, establishing topology layouts and efficient IP allocation schemes.
- **Configured enterprise network services** by setting up dedicated Web, Email, and DNS servers.

Air Quality Management System — Simulation & Hardware Prototyping
Industrial IOT based Academic Project
- **Individually designed an 8051-based air-quality management system from concept to simulation**, developing the initial system design on paper, assembling the circuit in NI Multisim, and writing the Assembly-language code to implement and test the system logic.
- **Verified the simulated system's operation** through functional testing and instructor assessment.
- **Collaborated on an ESP-based hardware prototype** integrating a smoke sensor as main air quality sensor, contributing to the code and hardware integration.

Modeling And Control Of Mechatronic Aero Pendulum
Control Systems based Academic Engineering Project
- **Designed the system concept and developed an initial physical prototype** for a mechatronic aero-pendulum intended to use feedback control for stabilizing and controlling its motion.
- **Explored mechanical fabrication and assembly**, working with local metalworking and carpentry resources to construct the frame and mount the system components.
- **Investigated sensor-integration and alignment challenges**, particularly the difficulty of mounting a potentiometer precisely enough to obtain reliable position feedback with the available tools and budget.
- **Evaluated practical implementation constraints** that prevented completion of the physical control system, including fabrication limitations and difficulties integrating the available feedback sensor into the prototype.

RF Wireless Data Transmission Model
DIgital Communications - Academic Project
- Designed and built a working miniature model using Arduino and 433-Mhz modules.

A Diagnostic Model Of Computer Systems In The Form Of Finite State Machine
Digital Logic Design - Research Implementation & Lab Project
- Utilized the Finite State Machine tool to develop a model of computer systems' states.

Online Attendance System
Data Structures and Algorithms - Academic Project
- Designed a simple yet functional online web app for students, instructor and coordinator for managing student attendances.

Wireless Voltage Detector
PCB Design - Collaborative Lab Project
- **Co-developed the conceptual design** within a three-member team, analyzing theoretical principles of non-contact voltage detection.
- **Simulated circuit behavior** to validate component selections, frequencies, and signal amplification before physical assembly.

- **Planned and executed the hardware fabrication**, soldering and assembling the physical prototype to achieve functional wireless detection.

Publications

Design and Analog Implementation of Synchronization Controllers for Feedback Instruments' Servo Trainer Modules 2024
International Conference on IT and Industrial Technologies (ICIT 2024) — IEEE

Syed Zulqarnain Haider, Muhammad Shafran, Dr. Umar Farooq, *Khurram Karim Qureshi, Muhammad Usman Asad and Jason Gu*

- **Paper overview -** Investigated the synchronization of two servo systems using analog proportional, proportional-integral, and state-feedback controllers. The controllers were implemented using operational amplifiers and breadboard circuitry, with experimental results demonstrating synchronized operation between master and slave servo systems.
- **My Contribution -** Contributed to the end-to-end research process, including system modeling, control-design calculations, MATLAB analysis, analog circuit design in Multisim, hardware implementation, and experimental testing. Prepared the main manuscript draft, technical documentation, and presentation materials, and independently learned basic LaTeX to format the manuscript in Overleaf for publication, incorporating feedback and revisions under faculty supervision.
- **DOI:** https://doi.org/10.1109/ICIT63607.2024.10860018

Relevant Technical Skills

Web Development

- **Frontend Architecture**: React, HTML5/CSS3, JavaScript (ES6+), TailwindCSS, Shadcn/ui
- **Visual Mechanics & Motion**: Framer Motion, Anime.js, tsParticles, GSAP (Fundamentals)
- **Cloud & Foundational Backend**: Supabase (PostgreSQL), Node.js, Express, MongoDB (Foundational)
- Design & User Flows**:** Figma, Google Stitch (**Design-First Prototyping Workflow**)
- **Workflow Automation**: AI-assisted rapid scaffolding, prompt-to-component translation, and automated library integration.

Technical Documentation & Project Coordination

- **Technical Writing:** Research manuscripts, engineering reports, technical documentation, project proposals, and presentation materials.
- **Documentation Tools:** Microsoft Word, basics of LaTeX/Overleaf.
- **Project Coordination:** Academic project planning, scheduling, milestone tracking, task coordination, and documentation.
- **Communication & Reporting:** Technical presentations, research revisions, and communicating project findings.

Programming

- **Languages**: JavaScript (ES6+), Python, C++, and Assembly (.asm).

Circuit Design, Embedded Systems & Engineering Prototyping

- **Circuit Design & Simulation -** NI Multisim, Proteus, schematic design, and circuit simulation.
- **3D CAD Design** - Basic part layout and foundational geometric modeling in **AutoCAD** and **FreeCAD**.
- **Mathematical Simulation -** Foundational engineering calculations, system modeling, and simulations in **MATLAB & Simulink.**
- **Embedded Hardware -** Arduino/ESP-based prototyping, microcontroller interfacing, sensor integration, breadboarding, and soldering.
- **Hardware Implementation -** Physical circuit assembly, testing, and basic troubleshooting.
- **Engineering Applications -** Circuit-level implementation of control systems, sensor-based systems, and wireless communication prototypes.

Network Systems

- **Network Engineering:** Designing topologies, configuring routers/switches, and mapping IP schemes in **Cisco Packet Tracer**.
- **Traffic Analysis:** Running basic packet captures and network diagnostics using **Wireshark**.

Achievements

Idea Presentation

ICECE, 2024 Idea Presentation/Poster Competition

- Presented a *concept for a smart, sustainable kitchen and miniature garden ecosystem,* **exploring integrated energy recovery, food-waste reuse, water reuse, and small-scale food cultivation.**

Positions In

SEECON ‘23 & ‘24 Matlab & Simulink Design Competitions, & Circuit Design

- Winner in SEECON '23 **Matlab Programming** Competition
- Runner Up in **Circuit Design and Simulink Design** Competitions in SEECON '23 and '24.

Courses & Certificates

Product School Micro-Certifications - Online
- Product Discovery (PDC) | 2026
- Product Strategy (PSC) | 2026
- Product Roadmaping (PRC) | 2026
- Product Launches (PRLC) | 2026
- Product Experimentation (PEC) | 2026
- Product Analytics (PAC) | 2026
- Product Prioritization (PPC) | 2026

Alison - Online
- Digital Project Management | 2026 (Completed, Unclaimed)

HP Life - HP Foundation - Online
- Design Thinking | 2026

Project Management Institute (PMI) - Online
- PMI Kickoff™ Agile Path | 2025
- PMI Kickoff™ Predictive Path | 2025

Additional Experience

Sports Event Operations & Initiative 2025
IEECE, University of the Punjab Lahore
- **Co-organized a four-day annual college sports event**, coordinating scheduling and operations across football and eSports activities.
- **Identified operational bottlenecks** during the event and developed a proposal for a multi-month intercollegiate league, outlining potential tournament-management infrastructure, RC engineering competitions, and supporting media workflows.

Relevant Coursework

- Entrepreneurship
- Electrical Network Analysis
- Data Structures and Algorithms
- Digital Signal Processing
- Digital Logic Design
- Linear Control Systems
- Engineering Management
- Wireless and Mobile Communications
- Programming Fundamentals
- Digital Communications
- Transducers & Instrumentation
- Technical Writing
- Applied Physics
- Transmission and Switching Systems
- Computer Communication Networks
- Embedded Systems

Languages

Balti — Urdu
English — Chinese

Personality Traits

- Diversely innovative and **systems-oriented** by nature, can be slow but diverse

- **Planning-focused** professional skilled in conceptualization, structuring ideas, and process mapping
- **Fast learner** with the ability to analyze complex concepts and develop long-term understanding
- I instinctively **connect the dots** between technical logic, empathetic design, and practicality to envision holistic solutions.
- **Collaborative team contributor** who stays focused on target when in teams but might go into recursively branched ideation when thinking alone
- Optimistic and **growth-oriented mindset** with a focus on continuous improvement and impact

Interests

Ideation
Think of real and unreal ideas

Philosophy
Ideas of different people & groups

Piano
Attempts on mobile phone

Research Market Trends
Looking at reports & websites of different companies

Anime
Watch character development, twists and politics

Workout & Martial Arts

Self Dev Books
Top Pick: Laws of Success - N. Hill

Pinterest & Instagram
For ideas & inspiration

Gaming
Playing FreeFire only to max ranks & level

Music
Listen to music & songs, & sometimes try to write (ft. chatbot)